

Testing the Portfolio Analysis Pack at a desk

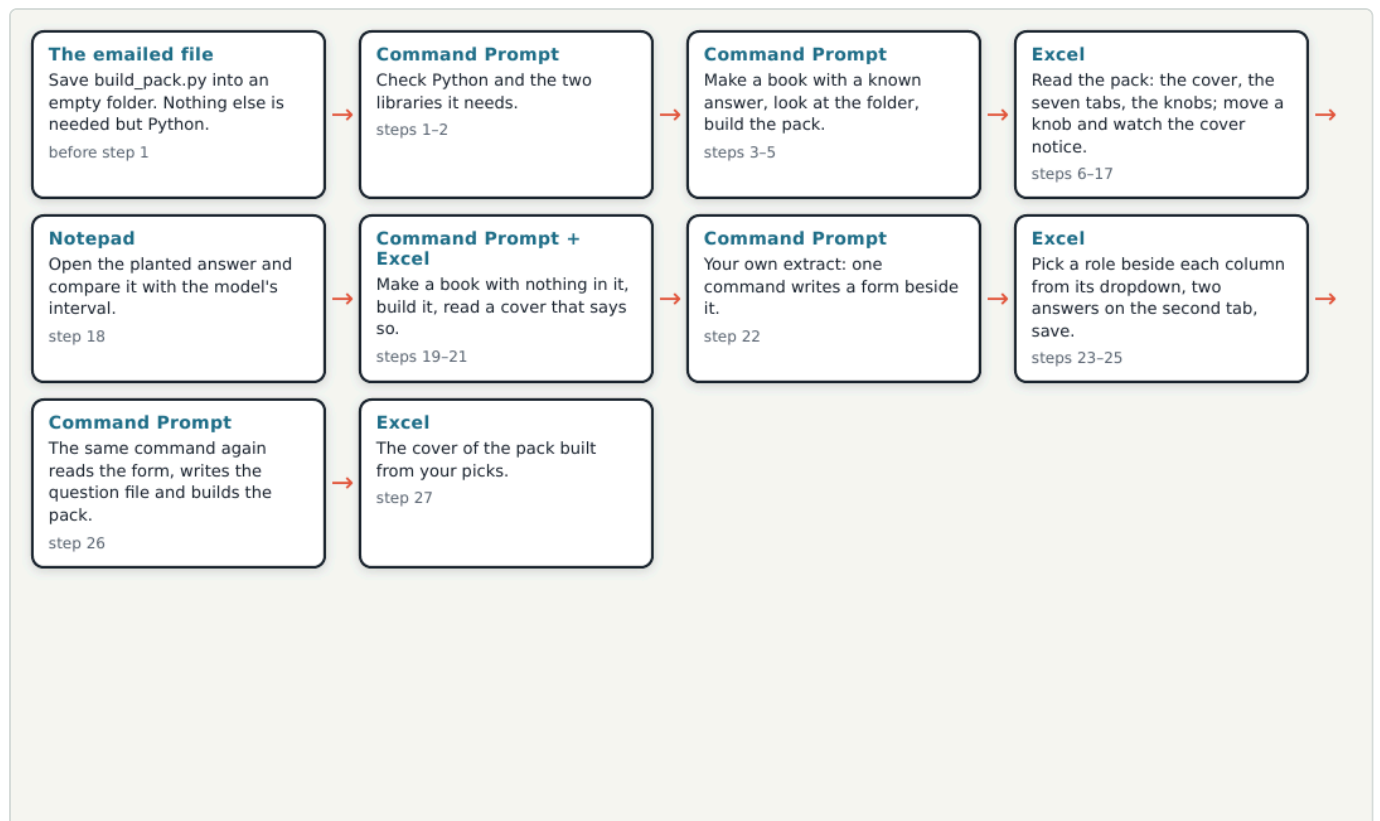
From the emailed file to a workbook you have read, in twenty-seven steps. Written for a person doing it for the first time, with a picture of every screen and what a correct one looks like.

This is the procedure for testing the pack on a machine that has never seen it: a file arrives by email, you run it on a book of made-up loans with a known answer, you read the workbook it builds, and then you point it at an extract of your own. It was walked four times on 22 September 2026: on the script as first written, which found eleven things the screens said wrong (`WALKTHROUGH-DEFECTS.md`); on the patched script, after the firm asked for everything wrong to be fixed before anything shipped; after Part E became a picker, so that nothing is edited by hand at the desk; and once more after the picker became a form in Excel, with a dropdown beside each column. Every screen below is from the last walk. Where a screen of yours differs from the picture, either the product has changed or something went wrong, and finding out which is the point of doing it again.

Two things to know before you start. The terminal pictures show a prompt that reads `desk>` ; on your machine it will show the folder you are in. And the spreadsheet pictures were taken with LibreOffice, because the machine the walk ran on has no Excel; the numbers and words are the same, the fonts are not.

You need: the emailed file `build_pack.py` , Python 3.10 or later, the two libraries it uses (`openpyxl` and `PyYAML` ; the script says which pip line to run if either is missing), and Excel. Nothing is edited by hand.

The route



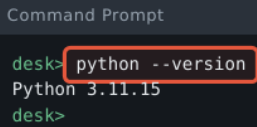
The route: ten screens in order, with what happens on each.

Make a new, empty folder, save `build_pack.py` into it, and open a Command Prompt in that folder (in File Explorer, type `cmd` in the address bar and press Enter). Every command below is typed there.

Part A · Is the machine ready?

Step 1 — Check Python

Do: type `python --version` and press Enter.



```
Command Prompt
desk> python --version
Python 3.11.15
desk>
```

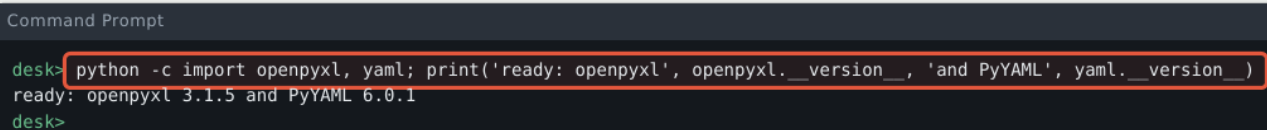
Step 1

Right when: it answers with a version of 3.10 or later. Here it was 3.11.15. If it says Python is not recognised, install it from python.org and tick "Add to PATH".

Step 2 — Check the two libraries

Do: type this exactly, on one line, and press Enter:

```
python -c "import openpyxl, yaml; print('ready: openpyxl', openpyxl.__version__, 'and PyYAML', yaml.__version__)"
```



```
Command Prompt
desk> python -c import openpyxl, yaml; print('ready: openpyxl', openpyxl.__version__, 'and PyYAML', yaml.__version__)
ready: openpyxl 3.1.5 and PyYAML 6.0.1
desk>
```

Step 2

Right when: it prints `ready:` with two version numbers. If it prints an error naming a module, run `pip install openpyxl PyYAML` and try again.

Part B · A book with a known answer

The script can make itself a book of 40,000 made-up loans in which a real effect has been planted, and write down what it planted. That is what you test on first: a book where you know the answer.

Step 3 — Make the book

Do: type `python build_pack.py --synth demo` and press Enter.

```
Command Prompt
desk> python build_pack.py --synth demo
wrote demo/loans.csv, config.yaml, planted.json (effect, seed 20260918, 40,000 loans)
then: python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
desk>
```

Step 3

Right when: the first line reads *wrote demo/loans.csv, config.yaml, planted.json (effect, seed 20260918, 40,000 loans)* and the second, *then:*, gives the build command, which is step 5's. Nothing else is printed. The planted answer is in `demo\planted.json`, and step 18 comes back to it.

Step 4 — Look at what appeared

Do: type `dir` (the picture shows `ls`, the same thing on the machine the walk ran on) and look at the folder.

```
Command Prompt
desk> ls -l . demo
.:
build_pack.py
demo

demo:
config.yaml
loans.csv
planted.json
desk>
```

Step 4

Right when: the folder holds `build_pack.py` and a `demo` folder with three files in it, and nothing else. The script unpacks itself into a temporary folder while it runs and removes it when it finishes.

Step 5 — Build the pack

Do: type this on one line and press Enter. It takes a few seconds.

```
python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
```

Command Prompt

```
desk> python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
wrote demo/pack.xlsx (165,150 bytes, sha256 908d6821cb7a46ee...)
read 0.2s · population 2.0s · models 1.6s · workbook 0.7s
29,343 seasoned loans, 257 events (event), 10,657 unseasoned excluded; gradient reads: monotonic increasing
step 4 size_band (edges): survives
step 4 amount_band (edges): survives
step 4 category_2 (levels): survives
step 6 M1: flag odds ratio 3.73 [2.84, 4.89] on 211 events over 11 coefficients, about 19 events per coefficient
step 6 M2: flag odds ratio 3.74 [2.85, 4.91] on 211 events over 21 coefficients, about 10 events per coefficient
formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1389 checks written to
_check
run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open demo/pack.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 5

Right when: the first line reads *wrote demo/pack.xlsx*, and the summary under it has these numbers: 29,343 seasoned loans, 257 events, *gradient reads: monotonic increasing*, three step-4 lines each ending *survives*, and M1's odds ratio 3.73 with the interval [2.84, 4.89] on 211 events, about 19 events per coefficient. The last three lines say that the formula check runs when Excel opens the file, that no run date was given so the pack will say so rather than guess one (add `--run-date 2026-09-22`, with the day you ran it, if you want it stamped), and what to do next. Keep this screen in mind; the cover in the next step must say the same things.

Part C · Read the workbook

Open `demo\pack.xlsx` in Excel. If Excel asks whether to enable editing or to recalculate, say yes. Read the tabs left to right; each of the next twelve steps is one tab or one thing on it.

Step 6 — The cover

Do: read the tab called *Cover*, top to bottom.

Portfolio Analysis Pack

synthetic_effect · run date not given · as-of 2026-06-30 · generator 0.1.0

The question

Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that real or something else in disguise?

The answer, in three lines

Gradient: yes. As the ratio rises, the event rate rises at every step.

Survives or collapses (step 4), for event: `size_band` (edges) — survives; `amount_band` (edges) — survives; `category_2` (levels) — survives; Model (step 6): the flag's odds ratio is 3.73 [2.84, 4.89] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added (M2) 211 events.

Regardless of the above

In 22.2% of the 25,106 seasoned loans where both `field_a` and `field_b` are present, `field_a + field_b > 1`.

`field_a`: decision one

`field_b`: decision two

Nothing in the process reacts to it.

What it rests on

29,343 seasoned loans, 257 events (event). 10,657 loans younger than 24 months are excluded from every rate.

1389 of 1389 formula checks agree (see `_check`)

1389 of 1389 formula checks agree (see `_check`)

the last line of the cover

Step 6

Right when: the black banner reads *run date not given*, because step 5 did not pass one. *The question* reads *Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that real or something else in disguise?* Under *The answer, in three lines* the three lines say what step 5's screen said: *Gradient: yes, survives three times (for event: `size_band` (edges) — survives; ...), and 3.73 [2.84, 4.89].* The last line reads *1389 of 1389 formula checks*

agree (see `_check`): every number Python wrote has been recomputed by the workbook's own formulas and matched. If that line shows a smaller first number, or the sentence about knobs having been moved (step 17 shows it), stop and say so.

Step 7 — Step 1, capture

Do: read the tab `1_Capture`.

Step 1 — Capture

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 1, capture. Before a field is trended it has to exist. For every origination quarter: loans originated, how many are still younger than the 24-month window at the as-of date, how many carry a blank in field_a, how many in field_b, and how many carry both. Blank is a state, not dirt: a quarter where a field was rarely captured is a finding about the field, and any trend in that quarter is a trend in capture, not in behaviour. Values the pack will not accept never reach this tab; the build refuses on them and lists the loans.

Quarter	Loans	Unseasoned	field_a blank	share	field_b blank	share	Both present	share
2019Q1	1,370	0	64	4.7%	610	44.5%	728	53.1%
2019Q2	1,370	0	78	5.7%	104	7.6%	1,194	87.2%
2019Q3	1,291	0	76	5.9%	99	7.7%	1,123	87.0%
2019Q4	1,386	0	74	5.3%	102	7.4%	1,210	87.3%
2020Q1	1,377	0	70	5.1%	121	8.8%	1,197	86.6%
2020Q2	1,237	0	79	6.4%	106	8.6%	1,057	85.4%
2020Q3	1,352	0	56	4.1%	132	9.8%	1,171	86.6%
2020Q4	1,306	0	57	4.4%	103	7.9%	1,147	87.8%
2021Q1	1,319	0	56	4.2%	107	8.1%	1,161	88.0%
2021Q2	1,327	0	70	5.3%	125	9.4%	1,140	85.9%
2021Q3	1,331	0	68	5.1%	105	7.9%	1,163	87.4%
2021Q4	1,337	0	70	5.2%	109	8.2%	1,168	87.4%
2022Q1	1,339	0	72	5.4%	108	8.1%	1,162	86.8%
2022Q2	1,342	0	75	5.6%	111	8.3%	1,166	86.9%
2022Q3	1,344	0	64	4.8%	98	7.3%	1,186	88.2%
2022Q4	1,342	0	66	4.9%	95	7.1%	1,185	88.3%
2023Q1	1,326	0	64	4.8%	111	8.4%	1,157	87.3%
2023Q2	1,284	0	78	6.1%	101	7.9%	1,115	86.8%
2023Q3	1,358	0	59	4.3%	112	8.2%	1,190	87.6%
2023Q4	1,311	0	59	4.5%	112	8.5%	1,149	87.6%
2024Q1	1,327	0	75	5.7%	114	8.6%	1,144	86.2%
2024Q2	1,367	0	72	5.3%	102	7.5%	1,198	87.6%
2024Q3	1,344	1,344	69	5.1%	109	8.1%	1,172	87.2%
2024Q4	1,283	1,283	77	6.0%	94	7.3%	1,117	87.1%
2025Q1	1,324	1,324	59	4.5%	94	7.1%	1,172	88.5%
2025Q2	1,370	1,370	72	5.3%	105	7.7%	1,205	88.0%
2025Q3	1,363	1,363	70	5.1%	112	8.2%	1,185	86.9%
2025Q4	1,379	1,379	58	4.2%	106	7.7%	1,220	88.5%
2026Q1	1,275	1,275	50	3.9%	94	7.4%	1,137	89.2%

Quarter	Loans	Unseasoned	field_a blank	share	field_b blank	share	Both present	share
2019Q1	1,370	0	64	4.7%	610	44.5%	728	53.1%
2019Q2	1,370	0	78	5.7%	104	7.6%	1,194	87.2%
2019Q3	1,291	0	76	5.9%	99	7.7%	1,123	87.0%
2019Q4	1,386	0	74	5.3%	102	7.4%	1,210	87.3%

the first four quarters, every column

Step 7

Right when: one row per origination quarter from 2019Q1, with the loans, how many are too young to count, and how many carry a blank in each field. The made-up book has a planted capture gap: 2019Q1 shows 610 loans (44.5%) with `field_b` blank, and every later quarter is under 10%. Quarters from 2024Q3 on show all their loans as unseasoned.

Step 8 — Step 2, prevalence

Do: read the tab 2_Prevalence.

Step 2 — Prevalence

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run data

Step 2, prevalence. Two series by origination quarter, on seasoned loans only, never merged: the capture rate is the share of seasoned loans with both field_a and field_b present; the flag rate is the share of those with both present where field_a + field_b > 1. Each carries a 95% Wilson interval. The span runs from 2019Q1 to 2024Q2: a change in the flag rate across those years is not on its own a read of underwriting drift, because the years a span covers can move either field for reasons of their own.

Quarter	Seasoned loans	Both present	Capture rate	Lower	Upper	Flagged	Flag rate	Lower	Upper
2019Q1	1,370	728	53.1%	50.5%	55.8%	155	21.3%	18.5%	24.4%
2019Q2	1,370	1,194	87.2%	85.3%	88.8%	250	20.9%	18.7%	23.3%
2019Q3	1,291	1,123	87.0%	85.0%	88.7%	262	23.3%	21.0%	25.9%
2019Q4	1,386	1,210	87.3%	85.4%	89.0%	251	20.7%	18.6%	23.1%
2020Q1	1,227	1,103	89.6%	87.7%	89.7%	258	21.5%	19.4%	24.3%
2020Q2	1,237	1,057	85.4%	83.4%	87.3%	244	23.1%	20.6%	25.7%
2020Q3	1,352	1,171	86.6%	84.7%	88.3%	265	22.6%	20.3%	25.1%
2020Q4	1,306	1,147	87.8%	85.9%	89.5%	273	23.8%	21.4%	26.4%
2021Q1	1,319	1,161	88.0%	86.2%	89.7%	237	20.4%	18.2%	22.8%
2021Q2	1,327	1,140	85.9%	83.9%	87.7%	248	21.8%	19.5%	24.2%
2021Q3	1,331	1,163	87.4%	85.5%	89.1%	263	22.6%	20.3%	25.1%
2021Q4	1,337	1,168	87.4%	85.5%	89.0%	252	21.6%	19.3%	24.0%
2022Q1	1,339	1,162	86.8%	84.9%	88.5%	293	25.2%	22.8%	27.8%
2022Q2	1,342	1,166	86.9%	85.0%	88.6%	265	22.7%	20.4%	25.2%
2022Q3	1,344	1,186	88.2%	86.4%	89.9%	263	22.2%	19.9%	24.6%
2022Q4	1,342	1,185	88.3%	86.5%	89.9%	269	22.7%	20.4%	25.2%
2023Q1	1,326	1,157	87.3%	85.4%	88.9%	251	21.7%	19.4%	24.2%
2023Q2	1,284	1,115	86.8%	84.9%	88.6%	250	22.4%	20.1%	25.0%
2023Q3	1,358	1,190	87.6%	85.8%	89.3%	246	20.7%	18.5%	23.1%
2023Q4	1,311	1,149	87.6%	85.8%	89.3%	263	22.9%	20.6%	25.4%
2024Q1	1,327	1,144	86.2%	84.2%	88.0%	240	21.0%	18.7%	23.4%
2024Q2	1,367	1,198	87.6%	85.8%	89.3%	277	23.1%	20.8%	25.6%

Quarter	Seasoned loans	Both present	Capture rate	Lower	Upper	Flagged	Flag rate	Lower	Upper
2019Q1	1,370	728	53.1%	50.5%	55.8%	155	21.3%	18.5%	24.4%
2019Q2	1,370	1,194	87.2%	85.3%	88.8%	250	20.9%	18.7%	23.3%
2019Q3	1,291	1,123	87.0%	85.0%	88.7%	262	23.3%	21.0%	25.9%
2019Q4	1,386	1,210	87.3%	85.4%	89.0%	251	20.7%	18.6%	23.1%

the first four quarters: capture rate and flag rate, each with its interval

Step 8

Right when: two rates per quarter, each with a lower and upper bound: the capture rate (both fields present) and the flag rate (the rule fires). Here the flag rate sits between 20% and 26% every quarter, and the table stops at 2024Q2, the last seasoned quarter.

Step 9 — Step 3, the gradient table

Do: read the tab *3_Gradient*, the table on the left.

Step 3 — Gradient

synthetic effect - source: loans.csv - sha256:19a1c61521c4ce9... - 29,343 seasoned loans - 257 events - generator 0.1.0 - run date not given

Step 3, gradient: Rate value = rate, a divided by field "rate" = 1, computed for the 29,343 seasoned loans where both fields are present (4,237 seasoned loans have a blank in one or both and are outside this table). Buckets are left-closed at the edges 0.5, 1, 2, 5; the bottom and top buckets are open. The base row is the 19,531 loans where the rule does not fire (field "a" = field "b" > 1). Each rate carries a 95% Wilson interval. Gap is the bucket rate minus the base rate in percentage points; multiple is the bucket rate divided by the base rate. Monotonic is real on the point estimates, comparing each pair of adjacent buckets that both hold loans. Columns headed chart: feed the chart beside the table and are not results.

Bucket	Loans	Events	Rate	Lower	Upper	Gap (pts)	Multiple	Rate change vs nearest bucket above with loans
< 0.5	12,536	39	0.31%	0.23%	0.42%	-0.22	0.59	
0.5 - 1	6,995	64	0.91%	0.72%	1.17%	0.39	1.73	0.0060
1 - 2	4,013	61	1.52%	1.19%	1.95%	0.99	2.88	0.0061
2 - 5	1,449	41	2.83%	2.09%	3.82%	2.30	5.37	0.0131
≥ 5	113	6	5.31%	2.46%	11.10%	4.78	10.07	0.0248
Base: rule does not fire	19,531	103	0.53%	0.44%	0.64%			

Monotonic?

monotonic increasing

Adjacent pairs whose intervals do not overlap

3

Interval clear of that bucket

chart bar up

chart bar down

chart bar rate

chart bar lower

chart bar upper

0.0011

0.0018

0.0034

0.0023

0.0042

0.0073

0.0117

0.0043

0.0033

0.0152

0.0119

0.0195

0.0099

0.0074

0.0283

0.0209

0.0382

0

0.0579

0.0386

0.0673

0.0346

0.1110

event: rate by bucket, with intervals

Outcome: event — 257 events among 29,343 seasoned loans

Bucket	Loans	Events	Rate	Lower	Upper	Gap (pts)	Multiple	Rate change vs nearest bucket above with loans
< 0.5	12,536	39	0.31%	0.23%	0.42%	-0.22	0.59	
0.5 - 1	6,995	64	0.91%	0.72%	1.17%	0.39	1.73	0.0060
1 - 2	4,013	61	1.52%	1.19%	1.95%	0.99	2.88	0.0061
2 - 5	1,449	41	2.83%	2.09%	3.82%	2.30	5.37	0.0131
≥ 5	113	6	5.31%	2.46%	11.10%	4.78	10.07	0.0248
Base: rule does not fire	19,531	103	0.53%	0.44%	0.64%			

Monotonic?

monotonic increasing

Adjacent pairs whose intervals do not overlap

3

the bucket table and the two lines under it

Step 9

Right when: five buckets of the ratio from < 0.5 to ≥ 5 and a base row. The rate climbs down the column: 0.31%, 0.91%, 1.52%, 2.83%, 5.31%. Under the table, *Monotonic?* reads *monotonic increasing* and *Adjacent pairs whose intervals do not overlap* reads 3. The two long headings wrap inside their columns. The columns headed *chart*: further right feed the chart and are not results; the tab's note says so.

Step 10 — Step 3, the chart

Do: on the same tab, look at the chart to the right.

Step 3 — Gradient

synthetic effect - source loans.csv - sha256 19a1c61521c4ce9... - 29,143 seasoned loans - 217 events - generator 0.1.0 - run date not given

Step 3, gradient: Rate value = rate, a divided by risk_rate = 1, computed for the 29,143 seasoned loans where both rates are present (4,237 seasoned loans have a blank in one or both and are outside this table). Buckets are left-closed at the edges 0.5, 1, 2, 5; the bottom and top buckets are open. The base row is the 18,531 loans where the rate does not fire (rate_a = 0, rate_b > 0). Each rate carries a 95% Wilson interval. gap is the bucket rate minus the base rate in percentage points; multiple is the bucket rate divided by the base rate. Monotonic is real on the point estimates, comparing each pair of adjacent buckets that both hold loans. Columns headed chart: feed the chart beside the table and are not results.

Outcome: event — 217 events among 29,143 seasoned loans

Bucket	Loans	Events	Rate	Lower	Upper	Gap (pts)	Multiple	Rate change vs nearest bucket above with loans	Interval clear of that bucket	chart: bar up	chart: bar down	chart: bar rate	chart: bar lower	chart: bar upper
< 0.5	12,536	39	0.31%	0.23%	0.42%	-0.22	0.59			0.0011	0.0008	0.0024	0.0023	0.0025
0.5 - 1	4,096	64	0.91%	0.72%	1.12%	0.39	1.73	6.0060	1	0.0026	0.0020	0.0063	0.0073	0.0077
1 - 2	4,013	61	1.52%	1.19%	1.95%	0.59	2.88	6.0061	1	0.0043	0.0033	0.0152	0.0119	0.0195
2 - 5	1,449	41	2.83%	2.09%	3.82%	2.30	5.37	6.0131	1	0.0099	0.0074	0.0283	0.0209	0.0482
> 5	113	6	5.31%	3.46%	11.30%	4.18	10.07	6.0318	0	0.0579	0.0385	0.0873	0.0346	0.1130
Base: rule does not fire	18,531	103	0.53%	0.44%	0.64%									

Monotonic?

monotonically increasing

Adjacent pairs whose intervals do not overlap

3

event: rate by bucket, with intervals

oper

042

117

195

382

110

event: rate by bucket, with intervals

the chart to the right of the table

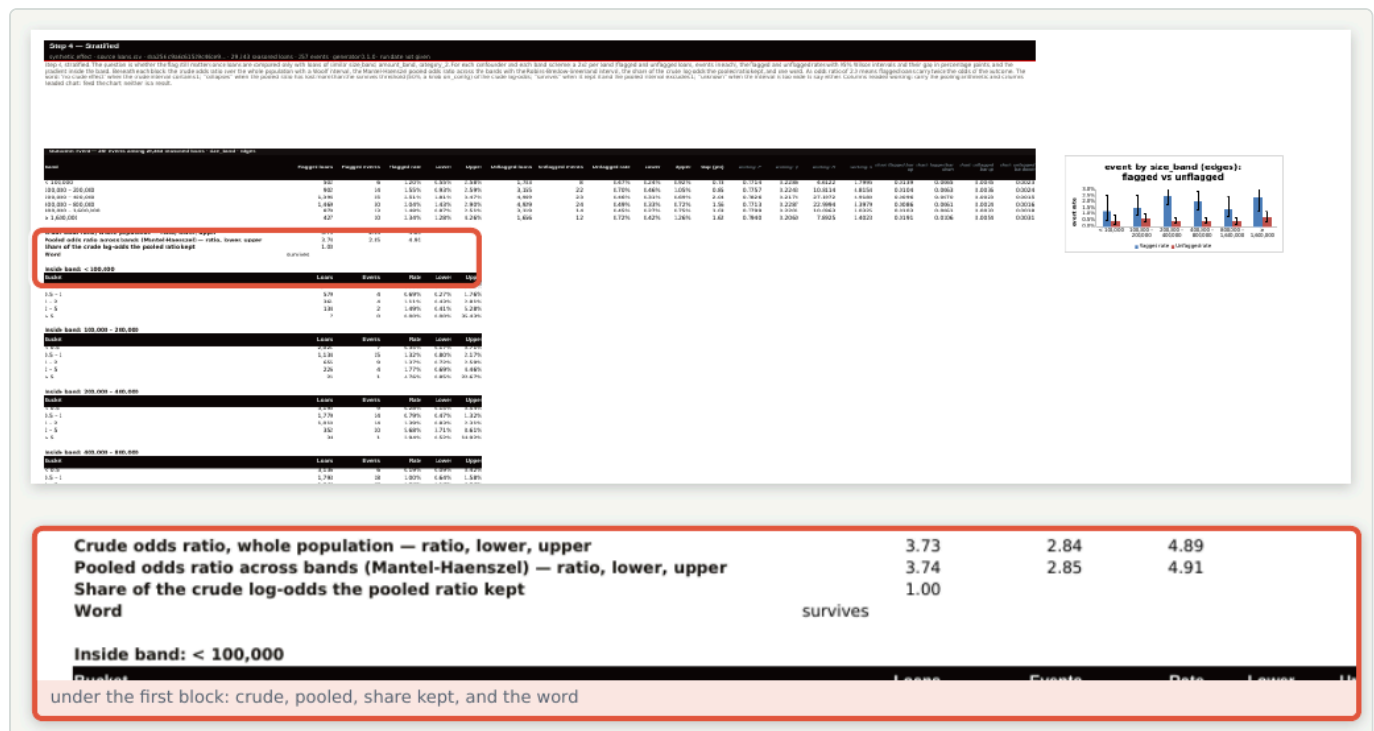
Step 10

Right when: five bars rising left to right, each with a whisker for its interval; the last bar's whisker is the widest, because that bucket holds only 113 loans.

Step 11 — Step 4, stratified

Do: read the tab 4_Stratified, the four short rows under the first block (they repeat

under every block).



Step 11

Right when: *Crude odds ratio, whole population — ratio, lower, upper 3.73, 2.84, 4.89; Pooled odds ratio across bands (Mantel-Haenszel) — ratio, lower, upper 3.74, 2.85, 4.91; Share of the crude log-odds the pooled ratio kept 1.00; and Word reads survives.* The labels read whole. The same word appears under all three blocks and matches the cover.

Step 12 — Step 5, decomposition

Do: read the tab `5_Decomposition`, the first block.

Step 5 — Decomposition

synthetic_effect · source: loans.csv · sha256:c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run_date not given

Step 5, decomposition. Where the flag's effect sits: for each dimension listed in the question file, the flagged and unflagged rates per level with 95% Wilson intervals, the gap in percentage points, the multiple, and the share of all flagged events that fall in the level. Rows are sorted by flagged events, most first, so a pattern that is all in one level is read off the top row. A dimension with a band scheme uses the scheme's first set of edges.

Outcome: event — 257 events among 29,343 seasoned loans · by category_1

category_1	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
south	1,118	28	2.50%	1.74%	3.60%	3,992	28	0.70%	0.49%	1.01%	1.80	25.9%
east	1,174	25	2.13%	1.45%	3.12%	3,960	16	0.40%	0.25%	0.66%	1.73	23.1%
north	1,095	20	1.83%	1.19%	2.80%	3,879	20	0.52%	0.33%	0.80%	1.31	18.5%
west	1,056	20	1.89%	1.23%	2.91%	3,870	17	0.44%	0.27%	0.70%	1.45	18.5%
central	1,132	15	1.33%	0.80%	2.17%	3,830	22	0.57%	0.38%	0.87%	0.75	13.9%

Outcome: event — 257 events among 29,343 seasoned loans · by amount_band

amount_band	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
60,000 - 250,000	3,483	66	1.95%	1.54%	2.47%	12,218	68	0.56%	0.44%	0.70%	1.40	63.0%
< 60,000	1,084	21	1.94%	1.27%	2.94%	3,737	15	0.40%	0.24%	0.66%	1.54	19.4%
≥ 250,000	1,008	19	1.88%	1.21%	2.93%	3,576	20	0.56%	0.36%	0.86%	1.33	17.6%

Outcome: event — 257 events among 29,343 seasoned loans · by origination_year

origination_year	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
2023	1,010	24	2.38%	1.60%	3.51%	3,601	13	0.36%	0.21%	0.62%	2.02	22.2%
2021	1,000	23	2.30%	1.54%	3.43%	3,632	16	0.44%	0.27%	0.71%	1.86	21.3%
2020	1,040	18	1.73%	1.10%	2.72%	3,527	31	0.88%	0.62%	1.24%	0.85	16.7%
2022	1,090	17	1.56%	0.93%	2.48%	3,609	18	0.50%	0.32%	0.79%	1.06	15.7%
2019	918	14	1.53%	0.91%	2.56%	3,337	14	0.42%	0.25%	0.70%	1.11	13.0%
2024	517	12	2.32%	1.33%	4.01%	1,825	11	0.60%	0.34%	1.08%	1.72	11.1%

Outcome: event — 257 events among 29,343 seasoned loans · by category_1

category_1	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
south	1,118	28	2.50%	1.74%	3.60%	3,992	28	0.70%	0.49%	1.01%	1.80	25.9%
east	1,174	25	2.13%	1.45%	3.12%	3,960	16	0.40%	0.25%	0.66%	1.73	23.1%
north	1,095	20	1.83%	1.19%	2.80%	3,879	20	0.52%	0.33%	0.80%	1.31	18.5%
west	1,056	20	1.89%	1.23%	2.91%	3,870	17	0.44%	0.27%	0.70%	1.45	18.5%
central	1,132	15	1.33%	0.80%	2.17%	3,830	22	0.57%	0.38%	0.87%	0.75	13.9%

the first dimension, rows sorted by flagged events

Step 12

Right when: one block per dimension listed in the question file, rows sorted by flagged events with the biggest first. For `category_1` the flagged rate sits between 1.3% and 2.5% in every row and the unflagged rate under 0.8%, so the effect is not hiding in one region.

Step 13 — Step 6, the model

Do: read the tab *6_Model*, the first row of the M1 table.

Step 6 — Model

synthetic_effect · source: loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 6, model. Two logistic regressions side by side, values only (a regression cannot be written as a cell formula, so this tab does not recalculate and its numbers have no check-tab twin). M1 has the flag plus the controls; M2 adds every confounder that is not already a control. A term's odds ratio says how the odds of the outcome move with that term, everything else held fixed: 2.0 on the flag means flagged loans carry twice the odds. The interval is the 95% range it plausibly sits in. If the flag's ratio falls from M1 to M2, what was lost is explained by the confounders. Categorical terms are measured against a reference level, the one with the most loans. A continuous control marked log enters as its natural logarithm. The tree beneath is a search for the two- or three-way combination with the most contrast, printed as rules with the loans, events and rate in each leaf; a lead to check, not a proof.

Outcome: event

Loans in the design: 25,106 seasoned loans with both rule fields and every predictor present; 0 left out for a blank predictor.

M1: flag + controls
25,106 loans · 211 events · 11 estimated coefficients · events per parameter 19.2

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.729	2.842	4.894	1.3162	0.1387	
origination_year=2021	1.164	0.736	1.843	0.1523	0.2343 against origination_year=2022	
origination_year=2023	1.104	0.694	1.758	0.0991	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M2: M1 + confounders not already controls
25,106 loans · 211 events · 21 estimated coefficients · events per parameter 10.0
Skipped: amount_band (its field amount is already a control)

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.742	2.851	4.911	1.3197	0.1387	
origination_year=2021	1.164	0.735	1.843	0.1519	0.2344 against origination_year=2022	
origination_year=2023	1.110	0.693	1.767	0.1042	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M1: flag + controls
25,106 loans · 211 events · 11 estimated coefficients · events per parameter 19.2

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.729	2.842	4.894	1.3162	0.1387	
origination_year=2021	1.164	0.736	1.843	0.1523	0.2343 against origination_year=2022	
origination_year=2023	1.104	0.694	1.758	0.0991	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M1: the flag row is the first term; its odds ratio and interval are the answer

Step 13

Right when: the row *flag* shows odds ratio 3.729, lower 2.842, upper 4.894, the same three numbers the cover rounds to two places. The line above the table reads *211 events · 11 estimated coefficients · events per parameter 19.2*; the M2 table below it says 3.742.

Step 14 — Step 7, the observation that stands regardless

Do: read the tab `7_Control`.

Step 7 — The observation that stands regardless

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 7, the observation that stands regardless of what steps 3 to 6 found. It rests on the counts alone: how often the two fields disagree in the seasoned book, and what the question file says reacts to that today. The tool asserts nothing of its own about how either field is verified or used; the two lines under the share are the question file's words.

Seasoned loans with both fields present
Of those, loans where the rule fires
Share

25,106
5,575
22.2%

In 22.2% of the 25,106 seasoned loans where both field_a and field_b are present, field_a + field_b > 1.
field_a: decision one
field_b: decision two
Nothing in the process reacts to it.

Seasoned loans with both fields present
Of those, loans where the rule fires
Share

25,106
5,575
22.2%

In 22.2% of the 25,106 seasoned loans where both field_a and field_b are present, field_a + field_b > 1.
field_a: decision one
field_b: decision two
Nothing in the process reacts to it.

the counts and the sentence built from them

Step 14

Right when: three counts (25,106 loans with both fields, 5,575 where the rule fires, share 22.2%) and the sentence built from them. This tab does not depend on anything steps 3 to 6 found.

Step 15 — The knobs

Do: read the tab `_config`, the block *Live knobs*.

Live knobs — change these and the workbook recalculates

Confidence level95.00%The interval width every rate carries. 0.95 means a 95% interval.
Interval methodWilsonWilson (default) or Clopper-Pearson. Pick from the list.
Survives threshold0.50Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.
z for the confidence level1.959964Derived from the confidence level; not a knob.

Rebuild knobs — shown for the record; change the question file and build again

Question filesynthetic_effect
Rulefield_a ÷ field_b > 1
Bucket edges0.5, 1, 2, 5
Outcomeevent, from outcome_date
Window (months)24
Filternone
As-of date2026-06-30

Live knobs recalculate the workbook when changed: the confidence level and the interval method. Rebuild knobs are shown for the record and take effect only when the pack is built again.

Live knobs — change these and the workbook recalculates

Confidence level95.00%The interval width every rate carries. 0.95 means a 95% interval.
Interval methodWilsonWilson (default) or Clopper-Pearson. Pick from the list.
Survives threshold0.50Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.
z for the confidence level1.959964Derived from the confidence level; not a knob.

the live knobs

Step 15

Right when: *Confidence level 95.00%, Interval method Wilson, Survives threshold 0.50* (its note: *Step 4's word: the share of the crude effect that must remain for 'survives'.* *0.50 means half.*), and *z 1.959964*. The block under it, *Rebuild knobs*, is for the record only; its *Outcome* line reads *event, from outcome_date*, the word and the column you gave.

Step 16 — Move a knob

Do: click the cell that reads *Wilson* and pick *Clopper-Pearson* from the list it offers. (In the walk, a script wrote the cell instead of a hand; the workbook does not know the difference.)

Live knobs — change these and the workbook recalculates

Confidence level

Interval method

Survives threshold

z for the confidence level

Clopper-Pearson

Wilson (default) or Clopper-Pearson. Pick from the list.

0.50

Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.

1.959964

Derived from the confidence level; not a knob.

Rebuild knobs — shown for the record; change the question file and build again

Question file

Rule

Bucket edges

Outcome

Window (months)

Filter

As-of date

synthetic_effect

field_a < field_b > 1

0.5, 1, 2, 5

event, from outcome_date

24

none

2026-06-30

Live knobs recalculate the workbook when changed: the confidence level and the interval method. Rebuild knobs are shown for the record and take effect only when the pack is built again.

Live knobs — change these and the workbook recalculates

Confidence level

Interval method

Survives threshold

z for the confidence level

95.00%

Clopper-Pearson

Wilson (default) or Clopper-Pearson. Pick from the list.

0.50

Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means hal

1.959964

Derived from the confidence level; not a knob.

Interval method now reads Clopper-Pearson

Step 16

Right when: the cell reads *Clopper-Pearson* and every interval in the workbook recalculates; the bounds on every tab move a little.

Step 17 — The cover notices

Do: go back to the *Cover* tab and read its last line.

Portfolio Analysis Pack
synthetic_effect, raw_data_not_generated - as-of 2026-06-30 - generator 2.1.0

The question
Loans where field_a divided by field_b is > 1: is event more common among them than among loans where it is not, and is that real or something else to be aware of?

The answer, in three lines
Gradient: yes. At the ratio level, the event rate rises at every step.
Survives or changes (step 4), for event size, band (edges) — survives; amount, band (edges) — survives; category, 2 (levels) — survives
Model (step 6): the flag's odds ratio is 2.72 [2.54, 4.28] with the controls (R1) and 2.74 [2.35, 4.91] with the confounders added (R2), or 2.11 events.

Agreement of the above
In 27.2% of the 26,326 unseasoned loans where both field_a and field_b are present, field_a > field_b > 1.

field_a: decision one
field_b: decision two
Nothing in the process leads to it.

What it rests on
26,326 unseasoned loans, 257 events (event), 10,457 loans younger than 24 months are excluded from every rate.

The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5). The Python column on _check was computed at these settings, so the formula check applies there only; set the knobs back to compare.

loans.csv - 0f4b1b1c-04b2-4273-b0c9-95074808c748 2026-06-30 10:00:00
synthetic_effect - 0f4b1b1c-04b2-4273-b0c9-95074808c748

The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5). The

the first half of the cover's last line, which now reads: The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5). The Python column on _check was computed at those settings, so the formula check applies there only; set the knobs back to compare.

Step 17

Right when: the line that read *1389 of 1389 formula checks agree* has been replaced by *The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5)...* and tells you to set the knobs back to compare. Set the knob back to Wilson and the original line returns. Close the workbook without saving.

Step 18 — The planted answer

Do: open `demo\planted.json` in Notepad.

demo\planted.json — Notepad

```
1 {
2   "seed": 20260918,
3   "loans": 40000,
4   "mode": "effect",
5   "effect_per_doubling": 2.0,
6   "base_rate": 0.012,
7   "asof": "2026-06-30",
8   "window_months": 24,
9   "true_marginal_flag_odds_ratio": 4.317157028211426,
10  "note": "true_marginal_flag_odds_ratio is the odds ratio of flagged (ratio > 1) against unflagged loans computed from the probabilities the generator drew with, before sampling noise and before seasoning; it is what the pack should recover within its interval."
11 }
```

Step 18

Right when: *true_marginal_flag_odds_ratio* is 4.317..., and that number lies inside the model's interval from step 13, 2.84 to 4.89. The note under it says this is what the pack should recover; it did.

Part D · A book with nothing in it

The second test is the opposite one: a book in which nothing was planted, where the pack must say so.

Step 19 — Make the empty book

Do: type `python build_pack.py --synth null --null` and press Enter.

```
Command Prompt
desk> python build_pack.py --synth null --null
wrote null/loans.csv, config.yaml, planted.json (null, seed 20260918, 40,000 loans)
then: python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
desk>
```

Step 19

Right when: the first line reads *wrote null/loans.csv, config.yaml, planted.json (null, seed 20260918, 40,000 loans)*.

Step 20 — Build it

Do: type this on one line and press Enter:

```
python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
```

```
Command Prompt
desk> python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
wrote null/pack.xlsx (164,930 bytes, sha256 e9e3b402e20775de...)
read 0.2s · population 1.6s · models 1.3s · workbook 0.7s
29,337 seasoned loans, 371 events (event), 10,663 unseasoned excluded; gradient reads: not monotonic
step 4 size_band (edges): no crude effect
step 4 amount_band (edges): no crude effect
step 4 category_2 (levels): no crude effect
step 6 M1: flag odds ratio 0.82 [0.62, 1.08] on 324 events over 11 coefficients, about 29 events per coefficient
step 6 M2: flag odds ratio 0.82 [0.62, 1.09] on 324 events over 21 coefficients, about 15 events per coefficient
formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1389 checks written to
_check
run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open null/pack.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 20

Right when: *gradient reads: not monotonic*, three step-4 lines each ending *no crude effect*, and M1's odds ratio 0.82 with an interval [0.62, 1.08] that has 1 inside it.

Step 21 — Its cover

Do: open `null\pack.xlsx` and read the three answer lines on the cover.

Portfolio Analysis Pack

synthetic_null · run date not given · as-of 2026-06-30 · generator 0.1.0

The question

Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that something else in disguise?

The answer, in three lines

Gradient: not one-directional. The event rate does not rise at every step of the ratio.

Survives or collapses (step 4), for event: `size_band` (edges) — no crude effect; `amount_band` (edges) — no crude effect; `category_2` (levels) — no crude effect

Model (step 6): the flag's odds ratio is 0.82 [0.62, 1.08] with the controls (M1) and 0.82 [0.62, 1.09] with the confounders added (M2) 324 events.

Regardless of the above

In 22.3% of the 25,119 seasoned loans where both `field_a` and `field_b` are present, $\text{field_a} \div \text{field_b} > 1$.

`field_a`: decision one

`field_b`: decision two

Nothing in the process reacts to it.

What it rests on

29,337 seasoned loans, 371 events (event). 10,663 loans younger than 24 months are excluded from every rate.

1389 of 1389 formula checks agree (see `_check`)

The answer, in three lines

Gradient: not one-directional. The event rate does not rise at every step of the ratio.

Survives or collapses (step 4), for event: `size_band` (edges) — no crude effect; `amount_band` (edges) — no crude effect; `category_2` (levels) — no crude effect

Model (step 6): the flag's odds ratio is 0.82 [0.62, 1.08] with the controls (M1) and 0.82 [0.62, 1.09] with the confounders added (M2) 324 events.

the three answer lines on the book with nothing planted

Step 21

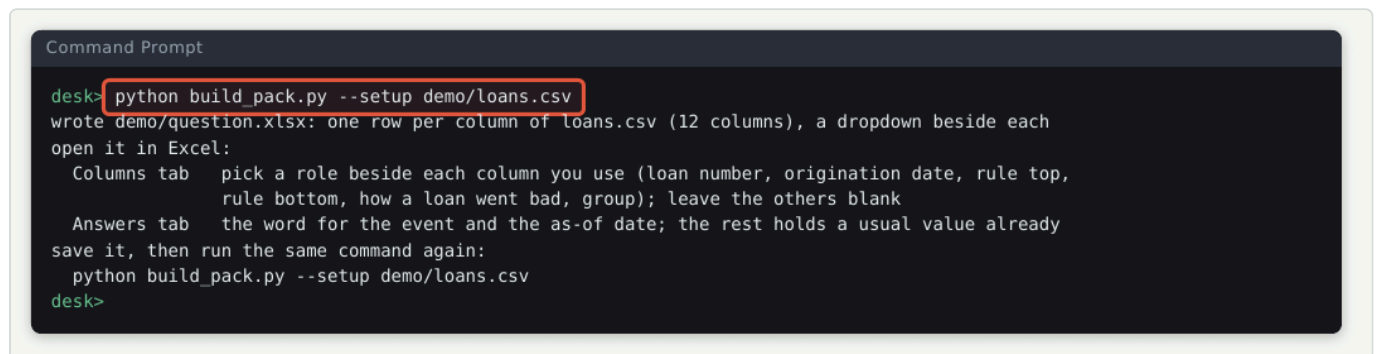
Right when: *Gradient: not one-directional, no crude effect three times, and 0.82 [0.62, 1.08].* A pack that found an effect here would be wrong. Close the workbook.

Part E · Your own extract

The made-up book came with its question file. An extract of your own does not, and the tool will not guess. One command writes a form beside your extract: one row per column, with what it reads as and three sample values, and a dropdown in the next cell for the column's role. You pick in Excel, save, and run the same command again; it reads the form, writes the question file from it, checks it, and builds the pack. Nothing is typed at a prompt and nothing is edited by hand. The walk did this on the made-up extract so the picks can be checked against Part B; on a real one the columns will be yours. (If you would rather answer at the keyboard, one question at a time, add `--ask` to the command.)

Step 22 — Ask for the form

Do: type `python build_pack.py --setup demo/loans.csv` and press Enter.



```
Command Prompt
desk> python build_pack.py --setup demo/loans.csv
wrote demo/question.xlsx: one row per column of loans.csv (12 columns), a dropdown beside each
open it in Excel:
  Columns tab  pick a role beside each column you use (loan number, origination date, rule top,
               rule bottom, how a loan went bad, group); leave the others blank
  Answers tab  the word for the event and the as-of date; the rest holds a usual value already
save it, then run the same command again:
python build_pack.py --setup demo/loans.csv
desk>
```

Step 22

Right when: *wrote demo/question.xlsx: one row per column of loans.csv (12 columns), a dropdown beside each*, then what to do on each tab, then the command to run again, spelled out. Nothing is built yet; the folder now holds `question.xlsx` beside the extract.

Step 23 — Open the form

Do: open `demo\question.xlsx` in Excel. Read the Columns tab: the number, the column, what it reads as, how often it is blank, three sample values, and beside them the three

cells to fill: *role*, *known only after the loan was made?*, and *band edges*.

loans.csv: which column is which

Pick a role beside each column you use; leave the others blank. Then the Answers tab, save, and run the command again.

#	column	reads as	blank	e.g.	role (pick from the list)	known only after the loan was made?	band edges (a group number column; smallest first)	its quartiles, from the data (used when the edges are blank)
1	loan_id	text	0%	L000001, L000002, L000003				
2	origination_date	date	0%	2022-06-15, 2025-06-24, 2015-09-14				
3	field_a	decimal	5%	203861.04, 306113.51, 708822.85				80000, 200000, 500000
4	field_b	decimal	9%	1022352.32, 389558.11, 474987.17				200000, 400000, 780000
5	amount	decimal	0%	711237.89, 70965.26, 93816.01				76000, 120000, 210000
6	category_1	text	0%	east, south, central				
7	category_2	text	0%	kind_6, kind_5, kind_3				
8	code_1	integer	0%	523567, 919149, 814961				310000, 520000, 720000
9	outcome_date	date	99%	2025-02-04, 2025-04-24, 2022-05-30				
10	flag_1	integer	0%	0, 1				0
11	measure_a	decimal	0%	261612.06, 43869.14, 10749.29				22000, 44000, 85000
12	measure_b	decimal	0%	711237.89, 70965.26, 93816.01				76000, 120000, 210000

one row per column of the extract; the shaded role cell beside each offers a list when clicked

Step 23

Right when: every column of the extract has its row (twelve here), the role cell of each is shaded and offers a list when you click it (loan number, origination date, rule top, rule bottom, rule top + group, rule bottom + group, went bad: date, went bad: flag, went bad: measure top, went bad: measure bottom, group), and the last column already shows the quartiles of each number column, from the data, for you to take or replace.

Step 24 — Pick a role beside each column you use

Do: from the dropdown, pick *loan number* beside `loan_id`, *origination date* beside `origination_date`, *rule top* beside `field_a`, *rule bottom + group* beside `field_b` (it is the bottom of the ratio and also the group that matters most), *went bad: date* beside `outcome_date`, and *group* beside `amount` and `category_2`. Leave the other five blank. In

the band-edges cell type 100000, 200000, 400000, 800000, 1600000 beside field_b and 60000, 250000 beside amount ; a group number column left blank takes the quartiles shown beside it.

loans.csv: which column is which

Pick a role beside each column you use; leave the others blank. Then the Answers tab, save, and run the command again.

#	column	reads as	blank	e.g.	role (pick from the list)	known only after the loan was made?	band edges (a group number column; smallest first)	its quartiles, from the data (used when the edges are blank)
1	loan_id	text	0%	L000001, L000002, L000003	loan number			
2	origination_date	date	0%	2022-06-15, 2026-06-24, 2025-09-14	origination date			
3	field_a	decimal	5%	203861.84, 306113.51, 708822.85	rule top			80000, 200000, 500000
4	field_b	decimal	9%	1022352.32, 389558.11, 474987.17	rule bottom + group		100000, 200000, 400000, 800000, 1600000	200000, 400000, 700000
5	amount	decimal	0%	711237.89, 70965.26, 93816.01	group		60000, 250000	70000, 120000, 210000
6	category_1	text	0%	east, south, central				
7	category_2	text	0%	kind_6, kind_5, kind_3	group			
8	code_1	integer	0%	523567, 919149, 814961				
9	outcome_date	date	99%	2025-02-04, 2025-04-24, 2022-05-30	went bad: date			310000, 520000, 720000
10	flag_1	integer	0%	0, 1				0
11	measure_a	decimal	0%	261612.06, 43869.14, 10749.29				22000, 44000, 85000
12	measure_b	decimal	0%	711237.89, 70965.26, 93816.01				70000, 120000, 210000

#	column	reads as	blank	e.g.	role (pick from the list)	known only after the loan was made?	band edges (a group number column; smallest first)	its quartiles, from the data (used when the edges are blank)
1	loan_id	text	0%	L000001, L000002, L000003	loan number			
2	origination_date	date	0%	2022-06-15, 2026-06-24, 2025-09-14	origination date			
3	field_a	decimal	5%	203861.84, 306113.51, 708822.85	rule top			80000, 200000, 500000
4	field_b	decimal	9%	1022352.32, 389558.11, 474987.17	rule bottom + group		100000, 200000, 400000, 800000, 1600000	200000, 400000, 700000
5	amount	decimal	0%	711237.89, 70965.26, 93816.01	group		60000, 250000	70000, 120000, 210000
6	category_1	text	0%	east, south, central				
7	category_2	text	0%	kind_6, kind_5, kind_3	group			
8	code_1	integer	0%	523567, 919149, 814961				
9	outcome_date	date	99%	2025-02-04, 2025-04-24, 2022-05-30	went bad: date			310000, 520000, 720000
10	flag_1	integer	0%	0, 1				0
11	measure_a	decimal	0%	261612.06, 43869.14, 10749.29				22000, 44000, 85000
12	measure_b	decimal	0%	711237.89, 70965.26, 93816.01				70000, 120000, 210000

the roles picked beside the seven columns in use, and edges typed for two of the groups

Step 24

Right when: seven roles picked, five cells blank, two sets of edges typed. A cell that does not match the list is refused by Excel as you type; a pick the tool cannot use (a text column as *rule top*, two loan numbers, a rule column marked as known only later) is refused at step 26, every problem at once with its cell.

Step 25 — The two answers that are not a column, then save

Do: on the Answers tab, type event beside The word every tab will use for the event and 2026-06-30 beside The as-of date the pack is built at. The other rows already hold a usual value (a ratio, fires above 1, steps 0.5, 1, 2, 5, 24 months, nothing reacts today);

change one only if you know better. Save the file and close it.

The answers that are not a column		
Two are required: the word for the event and the as-of date. The rest hold a usual value; change what you know better.		
question	answer	note
Compare the two rule columns as	ratio (top / bottom)	ratio, or difference (top - bottom)
The flag fires when the ratio (or difference) is above	0.5, 1, 2, 5	1 1 for a ratio; a difference usually 0
Steps of the ratio for the gradient (edges, smallest first, separated by commas)	event	For a difference, leave blank to take the quartiles of top - bottom from the data
The word every tab will use for the event (one word)	24	required, e.g. event or bad
Months a loan needs on book before it counts	2026-06-30	required; type it as a date, e.g. 2026-06-30
The as-of date the pack is built at		blank: the pack says run date not given
Today's date, to stamp on the pack		
What at the bank reacts to this contradiction today	none	
If a loan went bad by a flag: the value that means yes		
If a loan went bad by a measure: combine its two columns as	ratio (top / bottom)	1
If a loan went bad by a measure: bad at or above (one cut)		fill this or the edges below
If a loan went bad by a measure: or its band edges (smallest first)		

question	answer	note
Compare the two rule columns as	ratio (top / bottom)	ratio, or difference (top - bottom)
The flag fires when the ratio (or difference) is above	0.5, 1, 2, 5	1 1 for a ratio; a difference usually 0
Steps of the ratio for the gradient (edges, smallest first, separated by commas)	event	For a difference, leave blank to take the quartiles of top - bottom from the data
The word every tab will use for the event (one word)	24	required, e.g. event or bad
Months a loan needs on book before it counts	2026-06-30	required; type it as a date, e.g. 2026-06-30
The as-of date the pack is built at		blank: the pack says run date not given
Today's date, to stamp on the pack		
What at the bank reacts to this contradiction today	none	
If a loan went bad by a flag: the value that means yes		
If a loan went bad by a measure: combine its two columns as	ratio (top / bottom)	1
If a loan went bad by a measure: bad at or above (one cut)		fill this or the edges below
If a loan went bad by a measure: or its band edges (smallest first)		

the Answers tab: the two required answers filled (the word for the event, the as-of date); the rest as written

Step 25

Right when: the two shaded cells beside those questions are filled and the file is saved. Excel turns the date you typed into a date; that is fine, the tool reads either.

Step 26 — Run the same command again

Do: type `python build_pack.py --setup demo/loans.csv` again and press Enter.

```
Command Prompt
desk> python build_pack.py --setup demo/loans.csv
read demo/question.xlsx:
  loan number      loan_id
  origination date  origination_date
  rule              field_a / field_b, fires above 1, steps 0.5, 1, 2, 5
  went bad          outcome_date (the date it happened)
  groups            field_b (edges 100,000, 200,000, 400,000, 800,000, 1,600,000)
                   amount (edges 60,000, 250,000)
                   category_2 (level by level)
  window            24 months; as-of 2026-06-30; run date not given
  reacts today      none
wrote demo/question.yaml from question.xlsx
building demo/field_a_vs_field_b.xlsx ...
wrote demo/field_a_vs_field_b.xlsx (172,063 bytes, sha256 0bdfbc1106d78593...)
read 0.2s · population 1.6s · models 1.2s · workbook 0.8s
29,343 seasoned loans, 257 events (event), 10,657 unseasoned excluded; gradient reads: monotonic increasing
  step 4 field_b (edges): survives
  step 4 amount (edges): survives
  step 4 category_2 (levels): survives
  step 6 M1: flag odds ratio 3.72 [2.84, 4.88] on 211 events over 6 coefficients, about 35 events per coefficient
  step 6 M2: flag odds ratio 3.74 [2.85, 4.91] on 211 events over 18 coefficients, about 12 events per coefficient
formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1473 checks written to
_check
run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open demo/field_a_vs_field_b.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 26

Right when: *read demo/question.xlsx*: followed by one line per answer as the tool understood it (the loan number, the origination date, the rule `field_a / field_b`, fires above 1, steps 0.5, 1, 2, 5, *went bad outcome_date (the date it happened)*, the three groups with their edges, the window and as-of date, *run date not given*), then *wrote demo/question.yaml from question.xlsx, building demo/field_a_vs_field_b.xlsx ...* (the pack is named after the two columns), and the same summary as step 5: 29,343 seasoned loans, 257 events, *monotonic increasing*, three step-4 lines each ending *survives*, M1 3.72 [2.84, 4.88], and the *then:* line telling you to open the file in Excel. M1 has 6 coefficients where step 5 had 11: the form adds only the origination year as a control, where the made-up book's own file also holds the amount and a category. If the form was refused, the screen lists every problem with its cell (*Columns!F9, Answers!B7*); fix them in Excel, save, and run the command again.

Step 27 — Its cover

Do: open `demo\field_a_vs_field_b.xlsx` and read the three answer lines.

Portfolio Analysis Pack

`field_a_vs_field_b` · run date not given · as-of 2026-06-30 · generator 0.1.0

The question

Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that something else in disguise?

The answer, in three lines

Gradient: yes. As the ratio rises, the event rate rises at every step.

Survives or collapses (step 4), for event: `field_b` (edges) — survives; amount (edges) — survives; `category_2` (levels) — survives

Model (step 6): the flag's odds ratio is 3.72 [2.84, 4.88] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added 211 events.

Regardless of the above

In 22.2% of the 25,106 seasoned loans where both `field_a` and `field_b` are present, `field_a + field_b > 1`.

Nothing in the process reacts to it.

What it rests on

29,343 seasoned loans, 257 events (event). 10,657 loans younger than 24 months are excluded from every rate.

1473 of 1473 formula checks agree (see `_check`)

Source

`loans.csv` · sha256 c9a6c61529c46ce99b974ad0f8c74493a2c4e7f39c9948fbd4b45a19f5d9ae29 · 40,000 rows read · question f
`field_a_vs_field_b` · sha256 2065fa7649623c2d...

The answer, in three lines

Gradient: yes. As the ratio rises, the event rate rises at every step.

Survives or collapses (step 4), for event: `field_b` (edges) — survives; amount (edges) — survives; `category_2` (levels) — survives

Model (step 6): the flag's odds ratio is 3.72 [2.84, 4.88] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added (M2) 211 events.

the three answer lines of the pack built from your picks

Right when: *Gradient: yes, survives three times (for event: field_b (edges) — survives; amount (edges) — survives; category_2 (levels) — survives), and 3.72 [2.84, 4.88], with 1473 of 1473 formula checks agree lower down. The banner reads field_a_vs_field_b, the name the tool gave the question from your two columns, and run date not given. The same answer as step 6, from your own picks.*

To change a pick later, open `question.xlsx` again, change it, save, and run the command again; the question file is rewritten from the form each time.

When you are done

You have run the tool on a book with a known answer and watched it find the answer, on a book with no answer and watched it say so, and on an extract whose columns you designated yourself in Excel. To do the same on a real extract, run `python build_pack.py --setup YOUR-FILE.csv` (an `.xlsx` works too: add `--sheet` and the tab's name), fill the form it writes for that book (its own columns, its real as-of date, today's date to stamp, and the groups that matter for it), save, and run the command again. There is nothing to tidy afterwards: the script leaves nothing beside itself but the form, the question file and the pack.